

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

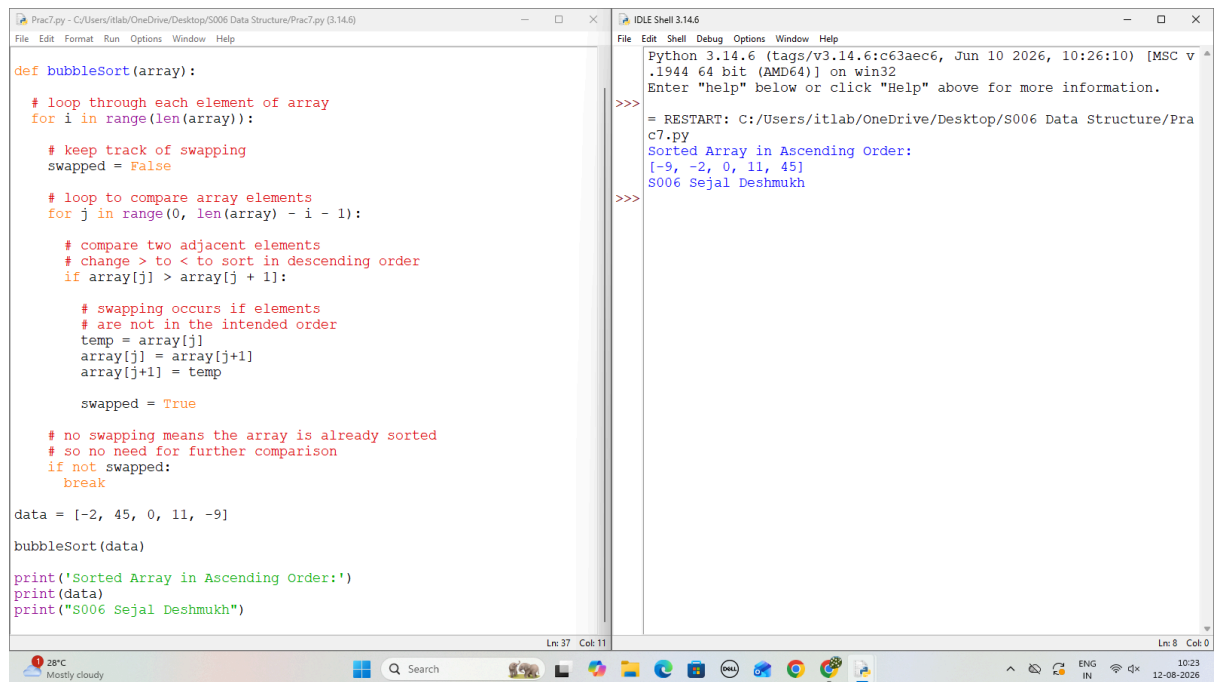
Practical - VIII

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Class :SYIT	Batch : 01
Date of Assignment :12/08/2026	Date/Time of Submission :12/08/2026

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort



```
Prac7.py - C:/Users/itlab/OneDrive/Desktop/S006 Data Structure/Prac7.py (3.14.6)
File Edit Format Run Options Window Help

def bubbleSort(array):
    # loop through each element of array
    for i in range(len(array)):
        # keep track of swapping
        swapped = False

        # loop to compare array elements
        for j in range(0, len(array) - i - 1):
            # compare two adjacent elements
            # change > to < to sort in descending order
            if array[j] > array[j + 1]:
                # swapping occurs if elements
                # are not in the intended order
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp

                swapped = True

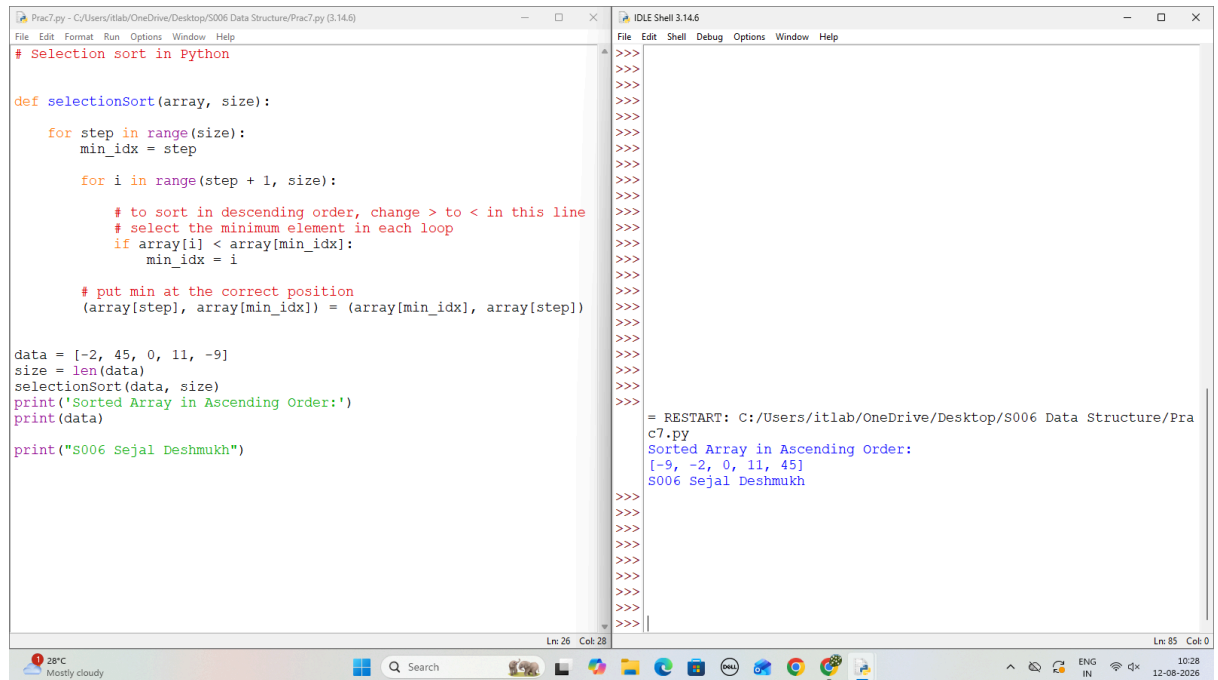
        # no swapping means the array is already sorted
        # so no need for further comparison
        if not swapped:
            break

data = [-2, 45, 0, 11, -9]
bubbleSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("S006 Sejal Deshmukh")

IDLE Shell 3.14.6
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v
.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> = RESTART: C:/Users/itlab/OneDrive/Desktop/S006 Data Structure/Pra
c7.py
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
S006 Sejal Deshmukh
>>>
```

b. Insertion sort



The screenshot shows a Python IDE with two windows. The left window, titled 'Prac7.py', contains the following code for Selection Sort:

```
# Selection sort in Python

def selectionSort(array, size):
    for step in range(size):
        min_idx = step
        for i in range(step + 1, size):
            # to sort in descending order, change > to < in this line
            # select the minimum element in each loop
            if array[i] < array[min_idx]:
                min_idx = i

            # put min at the correct position
            (array[step], array[min_idx]) = (array[min_idx], array[step])

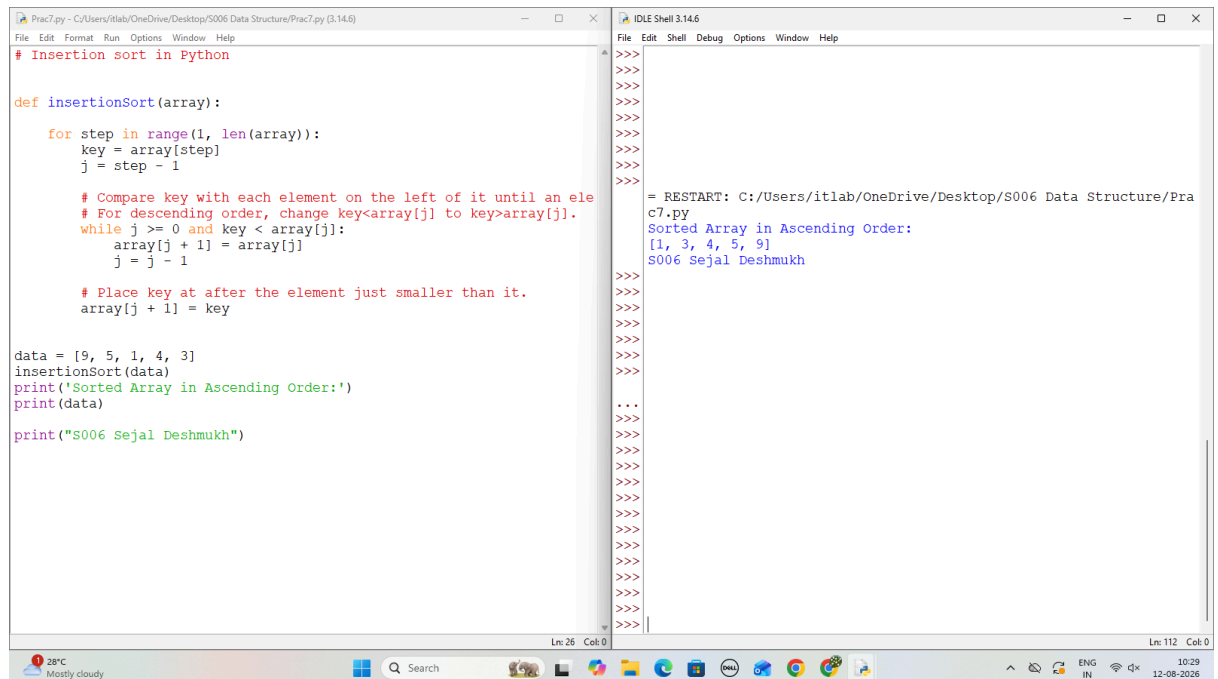
data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)

print("S006 Sejal Deshmukh")
```

The right window, titled 'IDLE Shell 3.14.6', shows the output of the program:

```
= RESTART: C:/Users/itlab/OneDrive/Desktop/S006 Data Structure/Pra
c7.py
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
S006 Sejal Deshmukh
```

c. Selection sort



The screenshot shows a Python IDE with two windows. The left window, titled 'Prac7.py', contains the following code for Insertion Sort:

```
# Insertion sort in Python

def insertionSort(array):
    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Compare key with each element on the left of it until an ele
        # For descending order, change key<array[j] to key>array[j].
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        # Place key at after the element just smaller than it.
        array[j + 1] = key

data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)

print("S006 Sejal Deshmukh")
```

The right window, titled 'IDLE Shell 3.14.6', shows the output of the program:

```
= RESTART: C:/Users/itlab/OneDrive/Desktop/S006 Data Structure/Pra
c7.py
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
S006 Sejal Deshmukh
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in descending order using **Bubble Sort**.
2. Write a Python program to sort a list of numbers in descending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in descending order using **Selection Sort**.

Theory Questions

4. What is sorting? Why is it required?
5. What is the basic idea of Bubble Sort?
6. How does Insertion Sort work?
7. How does Selection Sort work?
8. What is a **pass** in a sorting algorithm?
9. What is the difference between Bubble Sort and Selection Sort?
10. Which sorting algorithm is suitable when the data is already nearly sorted? Why?